DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, 91st Main Rd, 1st Stage, Kumaraswamy Layout,Bangalore 560078

DEPT. OF ELECTRONICS AND TELECOMMUNICATION ENGG

DigiKey

DERBI FOUNDATION ...Fueling Dreams

VERTEX

Sagar Hospitals® With You. For Life.

AXION

INNOVATE • BUILD • CONQUER

THREE CHALLENGES. ONE ENGINEERING ARENA.

BUILD FAST. THINK PRECISELY. PUT YOUR ENGINEERING INSTINCTS TO THE TEST.

12th & 13th October 2026

ET Department Building No. 17

01 / MOTION
VECTOR CHASE

02 / BUILD
HARDWARE EXPO

03 / LOGIC
EMBEDDED ENIGMA

SCAN TO REGISTER

Prize Pool
₹50,000+

Dr. B G Prasad
Principal

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Vertex Chief & Faculty Co-ordinator, Dept. of ETE

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GENERAL GUIDELINES

1. REGISTRATION

- Participants must register according to the team size specified for their event.
- All participants must be registered under their respective team.

2.IDENTIFICATION & REPORTING

- Participants must carry a valid college ID card for verification.
- Teams must report at the designated time as specified for their event.
- Late arrival will not be entertained.

3 .FAIR PLAY

- Participants are expected to maintain fair play and sportsmanship throughout the competition.
- Any form of cheating, unauthorized assistance, or violation of event rules may result in disqualification.

4 .CONDUCT

- Participants must follow the instructions of the organizing committee and event coordinators.
- Participants are expected to handle the venue, equipment, and event infrastructure responsibly.

5.DEVICES & COMMUNICATION

- Personal electronic devices may be restricted or prohibited during active competition rounds depending on the event.
- Communication with teams other than the registered teammates is not permitted during restricted competition phases.
- Where wireless systems are permitted, they must not interfere with venue infrastructure or other participants.

6. ORIGINALITY & AUTHORSHIP

- All projects, circuits, code, and builds must be created by the registered participants.
- Use of fully assembled, pre-built, or outsourced solutions is prohibited wherever the event requires original development.
- Participants must be able to explain their design, implementation, and technical decisions when questioned by the judges.

7. SAFETY

- All participants must follow organizer safety instructions throughout the event.
- Unsafe circuits, exposed mains connections, hazardous wiring, or builds that pose a risk to participants or equipment will not be permitted.
- Any participant found deliberately damaging the venue, equipment, or event infrastructure may be removed from the competition.

8.JUDGING & DISPUTES

- Judges will evaluate participants according to the criteria and event-specific rules.
- All required solutions, prototypes, or milestones must be verified by the organizing team where applicable.
- The decision of the judges and organizing committee regarding scoring, penalties, progression, and rule interpretation will be final and binding.

9. EVENT RULES

- Participants must follow all event-specific rules and regulations applicable to their event. In case of any clarification or dispute, the decision of the judges and organizing committee will be final and binding.

VECTOR CHASE

A high-speed autonomous challenge testing sensing, control, calibration, and real-time decisions. Strictly a line follower not a line maze solver.

TEAM SIZE
2–3 members

ENTRY FEE
₹100 per team

BOT SPECIFICATIONS

- Maximum dimensions: 25 cm × 25 cm × 25 cm
- A tolerance of ±2 cm is permitted during technical inspection.
- Maximum robot weight: 2.5 kg
- Onboard power supply only, with a maximum of 12 V DC
- The robot must be 100% autonomous.
- Bluetooth, Wi-Fi, RF, and manual external control are strictly prohibited.
- Only standard rubber wheels or treads are permitted.
- Adhesive tapes, sticky sprays, or chemicals that damage or adhere to the track are prohibited.

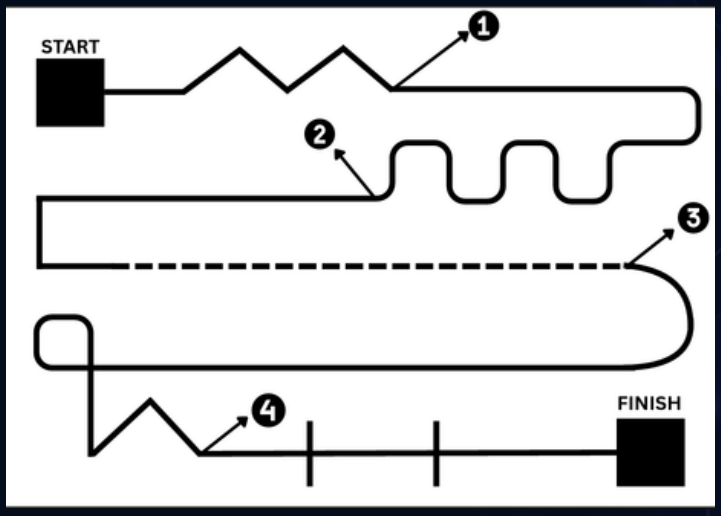
COMPONENT LOCK , TRIALS & TIMING

Once a bot clears technical inspection:

- Major components cannot be replaced with a different type, model, or specification.
- Replacement is permitted only with an identical model or exact equivalent specification.
- Major hardware changes between trials are prohibited.
- Each team receives 3 official trials.
- The team's best successful timing will be used for the final leaderboard.
- A 3-minute setup/calibration window is provided before the trial.
- Each trial has a maximum duration of 3 minutes.
- Timing begins when the front of the bot crosses the start line and ends when it crosses the finish line.
- Maximum 3 human touches/restarts per trial are permitted.
- After a touch, the bot must restart from the last successfully cleared checkpoint in the correct direction.

TRACK & ORIGINALITY

- The track will use non-reflective Flex Banner / Matte Poster material.
- Line thickness: 3 cm (30 mm).
- Both white-line-on-black and black-line-on-white sections may be used.
- The robot must dynamically handle line inversion.
- The track may include: Sharp grid intersections , T-junctions , Crossovers , U-shaped curves , C-shaped curves , S-shaped curves
- Participants must not step on, mark, fold, or damage the track.
- The robot must be designed, assembled, and programmed by the registered team. Readymade commercial bots and outsourced builds are prohibited. Judges may question participants about the circuit, code, and assembly; inability to explain the build may result in disqualification.



Note: This track is meant purely as a preview to give you an idea of what the actual track will be like. MANDATORY INSPECTION · Wireless control, outsourced bots, track shortcuts, arena damage, or unauthorized hardware changes may cause immediate disqualification.

EVENT COORDINATOR : Aditya Venkatesh · 9740768828

HARDWARE EXPO

Transform engineering concepts into functional physical prototypes with sound hardware integration and real-world applicability.

TEAM & REGISTRATION

2–4 members -No Entry Fee

TRACKS

01 INTELLIGENT SYSTEMS & AUTOMATION

- Robotics - Autonomous or remote-controlled systems, mobile robots, robotic arms, and custom mechatronics.
- Edge AI & TinyML On-device inference for gesture, sound, image recognition, and related embedded applications.
- Sustainable Tech Energy harvesting, automated irrigation, environmental monitoring, and embedded systems using communication protocols such as UART, I²C, SPI, or LoRa.

02 OPEN INNOVATION

- An open category for commercially viable engineering projects that fall outside the standard Track 1 domains while successfully bridging academic concepts with real-world applications.

ROUNDS

ROUND 1 · ONLINE SCREENING

- Teams must submit a 5-slide PDF presentation including Problem Statement, Hardware Block Diagram & Core Components
- Teams must submit a 1-minute working demonstration video
- Teams must also submit a complete Bill of Materials (BOM) with total project cost
- Shortlisting of top 20 teams is based on the team's development plan and working hardware

ROUND 2 · HARDWARE EXPO

- Top 20 teams advance to the in-person finale
- Teams must present a physical, working prototype
- Each team will be assigned a presentation table
- Judges will evaluate each team for approximately 5–10 minutes
- Evaluation includes:
  Live Demonstration , Hardware Integration Firmware

Note: Projects must strictly fall under at least one listed track.

HARDWARE EXPO

BUILD RULES

ORIGINAL BUILD & DEMONSTRATION

- Projects must be built from the ground up by the team.
- Arduino, ESP32, STM32, and similar development boards are permitted.
- Fully assembled plug-and-play kits are not permitted.
- Hardware must physically function during the live demonstration.
- Pure simulations or recordings of a currently non- functional project are not accepted.

INSPECTION & SAFETY

- Internal circuitry must remain accessible for inspection.
- Any team member may be asked to explain hardware architecture, firmware, control logic, and sensor processing.
- Wireless systems must not interfere with venue infrastructure or neighbouring projects.
- Exposed 220 V AC wiring is prohibited.
- High-current systems and large lithium battery packs must be wired safely.

JUDGING CRITERIA

Criterion	Weightage
Prototype Functionality	25%
Technical Complexity	25%
Build & Design Quality	20%
Innovation, Cost & Applicability	15%
Presentation & Pitch	15%

EVENT COORDINATOR : Dhruva J H · 6361536637

EMBEDDED ENIGMA

An intense four-round challenge from theoretical concepts to hands-on engineering. Teams solve unannounced puzzles, troubleshoot physical circuits, and decode mystery firmware against a strict clock.

TEAM SIZE
2 members

REGISTRATION FEE
₹150 per team

GENERAL RULES

COMPETITION PROTOCOL

- Use only the materials, components,and tools provided by the organizers.
- Outside hardware, pre-built circuits, and personal tools are not permitted in the event area.
- Phones, laptops, tablets, and smartwatches must remain stowed during active rounds.
- Teams may communicate only with their registered teammate.
- All challenges must be completed using the intended engineering process. Bypassing a challenge through physical or procedural shortcuts is not permitted.
- Component Responsibility: Treat all provided equipment with care.

VERIFICATION & CONDUCT

• An organizer must verify completed solutions before the team proceeds.
• Mishandling, destruction, or theft of equipment may lead to penalties or expulsion.
• Committee and judging decisions on scoring, progression, and rule interpretation are final and binding.

EVENT COORDINATOR: **Jyothika S · 8310325378**